

A Review on Large Language Models in Medical Imaging: Architectures, Applications, Taxonomies, Open Issues and Challenges

Abstract

The rapid advancement of Large Language Models (LLMs) has ushered in a new era in medical imaging, with these sophisticated artificial intelligence systems demonstrating remarkable capabilities in processing, understanding, and generating human-like text while increasingly integrating visual modalities. This comprehensive review examines the state-of-the-art applications of LLMs in medical imaging from 2015 to 2025, covering their architectural foundations, diverse clinical applications, taxonomic classifications, and the persistent challenges that must be addressed for successful clinical deployment. We systematically analyze how transformer-based architectures have evolved from text-only models to multimodal systems capable of processing medical images alongside clinical text, enabling unprecedented capabilities in diagnostic support, report generation, and clinical decision-making. While LLMs show tremendous promise in revolutionizing medical imaging workflows, significant challenges remain in areas of bias mitigation, explainability, data privacy, and clinical validation. This review provides a comprehensive framework for understanding the current landscape and future directions of LLMs in medical imaging.

Keywords: Large Language Models, Medical Imaging, Transformer Architecture, Multimodal AI, Clinical Decision Support, Computer-Aided Diagnosis

1. Introduction

The rapid advancement of artificial intelligence (AI) has ushered in a new era in natural language processing (NLP), with large language models (LLMs) like ChatGPT leading the way. Large language models (LLMs) are artificial intelligence (AI) tools specifically trained to process and generate text, and since the public release of ChatGPT in November 2022, these models have demonstrated capabilities nearly indistinguishable from human performance in many language tasks.

Medical images account for 90% of the data in digital medicine applications, making the integration of LLMs with medical imaging a critical area of research and development. Large language models in the medical and healthcare domain (LLMMs) have the potential to facilitate communication among healthcare professionals, patients, and their families, streamline the collection and analysis of patient health data, and assist in the development of surgical plans.

The intersection of LLMs and medical imaging represents a paradigm shift from traditional computer-aided diagnosis (CAD) systems toward more sophisticated, multimodal AI assistants capable of

understanding both visual and textual medical information. Multimodal data span a broad spectrum of data types, including medical images (eg, MRI and CT scans), time-series data (eg, sensor data from wearable devices and electronic health records), audio recordings (eg, heart and respiratory sounds and patient interviews), text (eg, clinical notes and research articles), videos (eg, surgical procedures), and omics data (eg, genomics and proteomics).

This review aims to provide a comprehensive analysis of LLMs in medical imaging, examining their architectural foundations, diverse applications, taxonomic classifications, and the challenges that must be addressed for successful clinical implementation.

2. Architectural Foundations of LLMs in Medical Imaging

2.1 Transformer Architecture Evolution

Language Transformer (Vaswani et al., 2017) is a neural network based on self-attention mechanisms and feed-forward module to compute representations and global dependencies. The transformer architecture, introduced in the seminal "Attention is All You Need" paper, has become the foundation for modern LLMs through its self-attention mechanism that enables efficient parallel processing and long-range dependency modeling.

2.2 Vision Transformers in Medical Imaging

The vision transformer (ViT) is a state-of-the-art architecture for image recognition tasks that plays an important role in digital health applications. Vision transformer as a feature extractor has the advantage of capturing the global context of the images effectively. In CNN-based feature extractor processes images hierarchically, extracting the local features only. Whereas ViT processes the image into small patches, and by projection, it will transform the patches into vector embedding allowing the dependencies more efficiently.

Key architectural variants include:

Vision Transformer (ViT): Our model utilizes transformer architectures, including Vision Transformer (ViT) and its variants like Data Efficient Image Transformer (DEiT) and BERT pre-training image transformer (BEiT), as an encoder.

Hybrid Architectures: Transformer, one of the latest technological advances of deep learning, has gained prevalence in natural language processing or computer vision. Since medical imaging bear some resemblance to computer vision, it is natural to inquire about the status quo of Transformers in medical imaging.

2.3 Multimodal Architecture Integration

M-LLMs hold transformative potential for numerous medical applications, demonstrating unparalleled proficiency in processing and integrating diverse data types. The evolution toward multimodal architectures represents a crucial advancement, enabling the integration of:

- **Vision-Language Models:** Vision-and-language models are a recent development, which take textual content and images as input and jointly learn to capture the relationships between both modalities
- **Cross-Modal Attention:** Our model utilizes a cross-attention mechanism between the vision transformer and GPT2, which enables it to create detailed and coherent medical reports
- **Unified Processing:** A transformer-based unified multimodal framework for Alzheimer's disease assessment

3. Taxonomy of LLM Applications in Medical Imaging

3.1 Primary Application Categories

Based on our systematic analysis, LLM applications in medical imaging can be taxonomically classified into the following categories:

3.1.1 Diagnostic Support and Image Interpretation

Medical Image Analysis:

- LLMs can also be trained to recognize and analyze medical images (e.g., X-rays, magnetic resonance imaging, and ultrasound), video, audio and remote photoplethysmograph signals to identify features and structures
- By analyzing the imaging data and considering the patient's medical history, these models can suggest potential diagnoses, differential diagnoses, and possible treatment options

Computer-Aided Diagnosis (CAD):

- Z. Zhao, S. Wang, J. Gu, Y. Zhu, L. Mei, Z. Zhuang, Z. Cui, Q. Wang, and D. Shen, Chatcad+: Towards a universal and reliable interactive CAD using LLMs
- Integration with existing diagnostic workflows

3.1.2 Report Generation and Documentation

Automated Report Generation: We propose an automated report generation model for medical images leveraging an encoder–decoder architecture. This includes:

- Radiology report generation
- Pathology report summarization

- Clinical documentation automation

Content Summarization:

- LLMs could output a formal discharge summary in a matter of seconds by analyzing all clinical notes
- Medical literature summarization
- Patient history synthesis

3.1.3 Clinical Decision Support

Treatment Planning: Rapidly expanding medical literature challenges oncologists seeking targeted cancer therapies. General-purpose large language models (LLMs) lack domain-specific knowledge, limiting their clinical utility

Risk Assessment:

- Models such as Gemini 1.5 Pro and Phi-3.5-instruct often score poorly for reasons other than performance per se
- Predictive modeling for clinical outcomes
- Patient stratification and triage

3.1.4 Education and Training

Medical Education:

- Radiologists, trainees, and even patients could interact with these models to obtain answers to questions related to radiology
- Simulation-based training
- Case-based learning platforms

3.2 Modality-Specific Applications

3.2.1 Radiology Applications

With the advent of large language models (LLMs), the artificial intelligence revolution in medicine and radiology is now more tangible than ever

Chest Imaging:

- O. Thawkar, A. Shaker, S. S. Mullappilly, H. Cholakkal, R. M. Anwer, S. Khan, J. Laaksonen, and F. S. Khan, XrayGPT: Chest radiographs summarization using medical vision-language models
- Pneumonia detection and COVID-19 diagnosis

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- Lung nodule characterization

Cross-sectional Imaging:

- CT and MRI interpretation
- Multi-planar reconstruction analysis
- 3D volume processing

3.2.2 Pathology Applications

The integration of large language models (LLMs) like ChatGPT in diagnostic medicine, with a focus on digital pathology, has garnered significant attention

Histopathology Analysis:

- Quilt-LLaVA: Visual Instruction Tuning by Extracting Localized Narratives from Open-Source Histopathology Videos
- Cancer grading and staging
- Morphological feature extraction

3.2.3 Specialized Imaging Modalities

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- Ophthalmology: Retinal imaging analysis
- Dermatology: Skin lesion classification
- Cardiology: Echocardiography interpretation

4. Technical Architectures and Methodologies

4.1 Pre-training Strategies

4.1.1 Foundation Model Development

Foundation models, including large language models (LLMs) and vision-language models (VLMs), have transformed the AI landscape, finding applications across diverse fields such as natural language processing, computer vision, and multimodal understanding

Medical-Specific Pre-training:

- L. Yang, Z. Wang, and L. Zhou, MedXChat: Bridging CXR modalities with a unified multimodal large model
- Domain-adapted vocabularies
- Medical literature corpus training

4.1.2 Transfer Learning Approaches

LLMs offer immense potential for enhancing various aspects of medical image processing. Additionally, the Transformer architecture, foundational to LLMs, is gaining prominence in this domain

4.2 Fine-tuning Methodologies

4.2.1 Task-Specific Adaptation

- This scoping review addresses this gap by examining studies on medical transformer-based language models and categorizing them into 6 tasks: dialogue generation, question answering, summarization, text classification, sentiment analysis, and named entity recognition
- Parameter-efficient fine-tuning
- Low-rank adaptation (LoRA)

4.2.2 Multimodal Training Strategies

The majority of existing research studies have addressed biomedical tasks using just one single data modality, however, modeling complex processes of biology and medicine inherently requires the integration of and learning on multiple modalities

4.3 Evaluation Frameworks

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4.3.1 Performance Metrics

The primary objective was accuracy, measured by whether the final diagnosis was within the LLM-generated ddx at the ICD-10 category level

Clinical Accuracy Measures:

- Diagnostic accuracy
- Sensitivity and specificity
- AUROC and AUPRC

Human Evaluation:

- Expert clinician assessment
- Inter-rater reliability
- Clinical utility scores

4.3.2 Benchmark Datasets

We evaluated six large language models of varying sizes and architectures in a zero-shot setting, meaning no additional fine-tuning was applied for any specific benchmark

5. Open Issues and Challenges

5.1 Bias and Fairness Issues

5.1.1 Data Bias Concerns

Language models, especially LLMs that are trained on large corpus, usually have intrinsic bias issues, since the training corpora often contains societal biases that are built into the model

Types of Bias:

- Bias from over-representativeness and under-representativeness: Certain groups, such as gender, age, race, religion, ethnicity, culture, political, or socioeconomic classes, may be under-represented or over-represented in the corpus
- Spatial and Temporal Bias: LLMs trained predominantly on the corpus from certain countries or geographic locations may absorb the cultural norms and values

5.1.2 Algorithmic Fairness

Deep learning algorithms have demonstrated remarkable efficacy in various medical image analysis (MedIA) applications. However, recent research highlights a performance disparity in these algorithms when applied to specific subgroups, such as exhibiting poorer predictive performance in elderly females

Mitigation Strategies:

- Bias mitigation post-processing for individual fairness
- Data augmentation techniques
- Fairness-aware training procedures

5.2 Explainability and Interpretability

5.2.1 Black Box Problem

Transparency in Machine Learning (ML), often also referred to as interpretability or explainability, attempts to reveal the working mechanisms of complex models

Technical Challenges:

- However, the variability in expert opinions can lead to biased and subjective assessments. Meanwhile, existing functionality-grounded evaluations still depend on manual annotations

- Complex attention mechanisms
- Multi-layer reasoning processes

5.2.2 Clinical Explainability Requirements

Following human-centered design principles in highly specialized and high stakes domains, such as medical image analysis, is challenging due to the limited access to end users and the knowledge imbalance between those users and ML designers

Implementation Approaches:

- The development of integrated explanation frameworks specifically designed for medical M-LLMs that can integrate both local and global explanations
- Visual attention maps
- Natural language explanations

5.3 Clinical Validation and Deployment Challenges

5.3.1 Regulatory Hurdles

Using language models for diagnostic medicine raises ethical and legal concerns. Patient privacy, data security, liability, and accountability are important considerations when deploying these models in clinical settings

Key Considerations:

- FDA approval processes
- Clinical trial requirements
- Post-market surveillance

5.3.2 Clinical Integration Challenges

While LLMs have achieved excellent performance on medical licensing exams, these tests fail to assess many skills that are necessary for deployment in a realistic clinical decision making environment, including gathering information, adhering to established guidelines, and integrating into clinical workflows

Workflow Integration:

- We find that models sometimes match the modalities requested by the doctors in the dataset, but often come to a diagnosis without requesting an abdominal imaging scan
- Electronic health record integration

- Real-time decision support systems

5.4 Technical Limitations

5.4.1 Computational Requirements

However, their high hardware requirements are being addressed through technological advancements

Resource Constraints:

- GPU memory requirements
- Inference latency
- Energy consumption

5.4.2 Data Quality and Availability

Limited Interpretation: Language models primarily provide text-based outputs and may not have the ability to interpret visual information, such as complex digital pathology images

Data Challenges:

- Medical image annotation quality
- Privacy-preserving data sharing
- Multi-institutional datasets

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5.5 Ethical and Privacy Considerations

5.5.1 Patient Privacy

Challenges include ensuring their empirical reliability, addressing ethical and societal implications, especially data privacy, and mitigating biases while maintaining privacy and accountability

Privacy Protection Measures:

- Differential privacy
- Federated learning approaches
- Data anonymization techniques

5.5.2 Informed Consent and Autonomy

There is a risk of overreliance on language models, potentially leading to reduced critical thinking or independent decision-making by healthcare professionals

6. Current State of the Art and Recent Developments

6.1 Leading Models and Frameworks

6.1.1 Commercial LLMs in Healthcare

Using a curated dataset of 2,400 cases and a framework to simulate a realistic clinical setting, current large language models are shown to incur substantial pitfalls when used for autonomous clinical decision-making

Major Players:

- GPT-4 and GPT-4V (OpenAI)
- In 2023, OpenAI released GPT-4, an M-LLM with the dual capability to process and respond to both text and images
- Google Med-PaLM and Med-Gemini
- Claude (Anthropic)

6.1.2 Open-Source Alternatives

Several open-source LLMs such as LLaMA, Flan-T5, Vicuna, and Alpaca have substantially driven progress and contributed to the field of LLMs

6.2 Multimodal Integration Advances

6.2.1 Vision-Language Models

The future of multimodal artificial intelligence models for integrating imaging and clinical metadata

Key Developments:

- MedXChat: Bridging CXR Modalities with a Unified Multimodal Large Model
- ECAMP: Entity-centered Context-aware Medical Vision Language Pre-training
- Cross-modal attention mechanisms

6.2.2 Specialized Medical Architectures

The advent of Large Language Models (LLMs) has sparked considerable interest in the medical image domain, as they can generalize to multiple tasks and offer outstanding performance

6.3 Performance Benchmarks and Evaluation

6.3.1 Diagnostic Accuracy Achievements

The relatively low accuracy of all 3 models at the ICD-10 category level, coupled with a mean of >3 out of 5 diagnoses located in a chapter outside the final diagnosis chapter, collectively suggest either a

knowledge or reasoning gap in current LLMs

6.3.2 Clinical Trial Results

- Multi-center validation studies
- Comparative effectiveness research
- Real-world evidence generation

7. Future Directions and Research Opportunities

7.1 Technological Advancement Opportunities

7.1.1 Architecture Improvements

The potential applications of LLMs in clinical settings are promising, with far-reaching implications for future research, clinical practice, and healthcare policy

Emerging Trends:

- Mixture of experts (MoE) architectures
- Retrieval-augmented generation (RAG)
- Few-shot learning capabilities

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7.1.2 Multimodal Integration Enhancement

We unify the discussion by focusing on how M-LLMs can serve as a transformative tool that integrates various data modalities to enhance health care outcomes

7.2 Clinical Implementation Pathways

7.2.1 Regulatory Framework Development

- Adaptive clinical trial designs
- Real-world evidence protocols
- Post-market monitoring systems

7.2.2 Healthcare System Integration

Future research should focus on the ethical integration of LLMs into clinical workflows. This requires robust data privacy safeguards, XAI to build trust, and transparency in MTB deliberations

7.3 Research Priorities

7.3.1 Validation Studies

understanding the challenges and barriers associated with the use of LLMs in this context is crucial for their successful implementation

Critical Research Areas:

- Large-scale clinical trials
- Health economic evaluations
- Patient safety monitoring

7.3.2 Technical Innovation

- Federated learning implementations
- Edge computing deployment
- Privacy-preserving techniques

8. Conclusion

This comprehensive review has examined the rapidly evolving landscape of Large Language Models in medical imaging, spanning architectural innovations, diverse applications, taxonomic classifications, and persistent challenges from 2015 to 2025. The transformative impact of AI in medical image processing is undeniable, and its continued development and implementation are poised to reshape the healthcare landscape for the better.

Key findings from our analysis include:

1. **Architectural Evolution:** The progression from text-only LLMs to sophisticated multimodal systems capable of processing both medical images and clinical text represents a fundamental shift in AI capabilities for healthcare.
2. **Application Diversity:** LLMs have demonstrated utility across the full spectrum of medical imaging applications, from automated report generation to complex diagnostic support and clinical decision-making.
3. **Persistent Challenges:** Despite remarkable technological advances, significant challenges remain in bias mitigation, explainability, clinical validation, and ethical deployment.
4. **Future Potential:** LLMs hold promise for transformative impacts in health care, with emerging capabilities in multimodal reasoning, few-shot learning, and clinical workflow integration.

The successful integration of LLMs into medical imaging practice will require continued collaboration between AI researchers, clinicians, regulators, and healthcare institutions. Collaborative Model Development: Collaboration between AI developers, healthcare professionals, and domain experts is crucial for the successful integration of LLMs in diagnostic medicine.

As we move forward, the focus must shift from purely technological achievements to comprehensive clinical validation, ensuring that these powerful tools enhance rather than replace human expertise while maintaining the highest standards of patient safety and care quality. The next decade will likely witness the maturation of LLM technology in medical imaging, transitioning from research demonstrations to routine clinical practice, fundamentally transforming how medical imaging is practiced globally.

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